



Ohio State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Ohio's Learning Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Ohio's Learning Standards for Mathematics to fully support Ohio's students and teachers. The tables on the following pages, organized by Ohio Learning Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
K.CC.1	Count to 100 by ones and by tens.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lessons 15-18
K.CC.2	Count forward within 100 beginning from any given number other than 1.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lessons 16-18
K.CC.3	Write numerals from 0 to 20. Represent a number of objects with a written numeral 0-20 (with 0 representing a count of no objects).	Mission 1, Topic D, Lessons 12-16 Mission 1, Topic E, Lessons 17-22 Mission 1, Topic F, Lessons 24-28 Mission 1, Topic G, Lessons 29-32 Mission 1, Topic H, Lessons 33- 37 Mission 4, Topic A, Lessons 1-6 Mission 4, Topic B, Lessons 7-12 Mission 4, Topic E, Lessons 25, 27 Mission 5, Topic A, Lessons 1-4 Mission 5, Topic B, Lessons 6, 9 Mission 5, Topic C, Lessons 10-13
K.CC.4	Understand the relationship between numbers and quantities; connect counting to cardinality using a variety of objects including pennies.	Mission 1, Topic B, Lessons 5-6 Mission 1, Topic C, Lessons 7-11 Mission 1, Topic D, Lessons 14-16 Mission 1, Topic E, Lessons 17-22 Mission 1, Topic F, Lessons 23-28 Mission 1, Topic G, Lessons 29-32 Mission 1, Topic H, Lessons 33-38
K.CC.5	Count to answer “how many?” questions about as many as 20 things arranged in a line, a rectangular array, or a circle, or as many as 10 things in a scattered configuration; given a number from 1-20, count out that many objects.	Mission 1, Topic C, Lessons 8-11 Mission 1, Topic D, Lessons 12-16 Mission 1, Topic E, Lessons 17-22 Mission 1, Topic F, Lessons 23-28 Mission 1, Topic G, Lessons 29-32 Mission 1, Topic H, Lessons 33-37 Mission 3, Topic F, Lessons 23-24 Mission 5, Topic A, Lessons 1-5 Mission 5, Topic B, Lesson 6 Mission 5, Topic C, Lessons 10-14 Mission 5, Topic D, Lessons 18, 19

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 5, Topic E, Lesson 22
K.CC.6	Orally identify (without using inequality symbols) whether the number of objects in one group is greater/more than, less/fewer than, or the same as the number of objects in another group, not to exceed 10 objects in each group.	Mission 3, Topic E, Lessons 17-19 Mission 3, Topic F, Lessons 20-24 Mission 3, Topic G, Lessons 25-28
K.CC.7	Compare (without using inequality symbols) two numbers between 0 and 10 when presented as written numerals.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lessons 25-28
Operations & Algebraic Thinking		
K.OA.1	Represent addition and subtraction with objects, fingers, mental images, drawings, sounds such as claps, acting out situations, verbal explanations, expressions, or equations. Drawings need not show details, but should show the mathematics in the problem. (This applies wherever drawings are mentioned in the Standards.)	Mission 4, Topic C, Lessons 13-16 Mission 4, Topic D, Lessons 20-24 Mission 4, Topic F, Lessons 29, 30 Mission 4, Topic G, Lesson 33 Mission 4, Topic H, Lesson 38
K.OA.2	Solve addition and subtraction problems (written or oral), and add and subtract within 10 by using objects or drawings to represent the problem.	Mission 4, Topic C, Lessons 16-18 Mission 4, Topic D, Lessons 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lessons 34-36 Mission 4, Topic H, Lesson 37
K.OA.3	Decompose numbers and record compositions for numbers less than or equal to 10 into pairs in more than one way by using objects and, when appropriate, drawings or equations.	Mission 1, Topic C, Lessons 9-11 Mission 1, Topic D, Lessons 14, 15 Mission 4, Topic A, Lessons 2, 4-6 Mission 4, Topic B, Lessons 7-12 Mission 4, Topic D, Lessons 22-24 Mission 4, Topic E, Lessons 25-28 Mission 4, Topic F, Lesson 32
K.OA.4	For any number from 1 to 9, find the number that makes 10 when added to the given number, e.g., by using objects or drawings, and record the answer with a drawing or, when appropriate, an equation.	Mission 4, Topic H, Lessons 39, 40
K.OA.5	Fluently add and subtract within 5.	Mission 4, Topic G, Lessons 34-36 Mission 4, Topic H, Lessons 37, 39

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 6, Topic A, Lessons 1, 2 Mission 6, Topic B, Lesson 5
Numbers & Operations in Base 10		
K.NBT.1	Compose and decompose numbers from 11 to 19 into a group of ten ones and some further ones by using objects and, when appropriate, drawings or equations; understand that these numbers are composed of a group of ten ones and one, two, three, four, five, six, seven, eight, or nine ones.	Mission 5, Topic A, Lessons 2-5 Mission 5, Topic B, Lessons 6-9 Mission 5, Topic C, Lessons 11-13 Mission 5, Topic E, Lessons 20, 21, 23
Measurement & Data		
K.MD.1	Identify and describe measurable attributes (length, weight, and height) of a single object using vocabulary terms such as long/short, heavy/light, or tall/short.	Mission 3, Topic A, Lessons 1, 2 Mission 3, Topic C, Lesson 8 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
K.MD.2	Directly compare two objects with a measurable attribute in common, to see which object has “more of”/“less of” the attribute, and describe the difference. For example, directly compare the heights of two children and describe one child as taller/shorter.	Mission 3, Topic A, Lessons 1-3 Mission 3, Topic B, Lessons 4-7 Mission 3, Topic C, Lessons 8-12 Mission 3, Topic D, Lessons 13-15 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lessons 29-32
K.MD.3	Classify objects into given categories; count the numbers of objects in each category and sort the categories by count. The number of objects in each category should be less than or equal to ten. Counting and sorting coins should be limited to pennies.	Mission 1, Topic A, Lessons 1-3 Mission 1, Topic B, Lessons 4-6 Mission 1, Topic C, Lesson 7 Mission 1, Topic G, Lesson 29 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7
Geometry		
K.G.1	Describe objects in the environment using names of shapes, and describe the relative positions of these objects using terms such as above, below, beside, in front of, behind, and next to.	Mission 2, Topic A, Lesson 5 Mission 2, Topic B, Lesson 8 Mission 2, Topic C, Lesson 10
K.G.2	Correctly name shapes regardless of their orientations or overall size.	Mission 2, Topic A, Lessons 2-5 Mission 2, Topic B, Lessons 7-8 Mission 2, Topic C, Lessons 9, 10

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.G.3	Identify shapes as two-dimensional (lying in a plane, "flat") or three-dimensional ("solid").	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 6 Mission 2, Topic C, Lessons 9, 10
K.G.4	Describe and compare two- or three-dimensional shapes, in different sizes and orientations, using informal language to describe their commonalities, differences, parts, and other attributes.	Mission 2, Topic A, Lessons 1-4 Mission 2, Topic B, Lessons 6, 7 Mission 2, Topic C, Lessons 9, 10 Mission 6, Topic A, Lesson 3
K.G.5	Model shapes in the world by building shapes from components, e.g., sticks and clay balls, and drawing shapes.	Mission 6, Topic A, Lessons 1-3
K.G.6	Combine simple shapes to form larger shapes.	Mission 6, Topic B, Lessons 5-7

1st Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Operations & Algebraic Thinking		
1.OA.1	Use addition and subtraction within 20 to solve word problems involving situations of adding to, taking from, putting together, taking apart and comparing, with unknowns in all positions, e.g., by using objects, drawings, and equations with a symbol for the unknown number to represent the problem.	Mission 1, Topic B, Lessons 4-8 Mission 1, Topic C, Lessons 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lessons 28-32 Mission 2, Topic A, Lessons 3-11 Mission 2, Topic B, Lessons 11-13, 17, 18, 21 Mission 2, Topic C, Lessons 22-24 Mission 2, Topic D, Lessons 27-29 Mission 3, Topic D, Lessons 10-13 Mission 4, Topic E, Lessons 19-22 Mission 6, Topic F, Lessons 25-27
1.OA.2	Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20, e.g., by using objects, drawings, and equations with a symbol for the unknown number to represent the problem. Drawings need not show details, but should show the mathematics in the problem. (This applies wherever drawings are mentioned in the Standards.)	Mission 2, Topic A, Lesson 1
1.OA.3	Apply properties of operations as strategies to add and subtract.3 Examples: If $8 + 3 = 11$ is known, then $3 + 8 = 11$ is also known. (Commutative property of addition.) To add $2 + 6 + 4$, the second two numbers can be added to make a ten, so $2 + 6 + 4 = 2 + 10 = 12$. (Associative property of addition.) Students need not use formal terms for these properties.	Mission 1, Topic E, Lessons 19-20 Mission 1, Topic F, Lessons 21-24 Mission 2, Topic A, Lessons 2, 6, 10, 11
1.OA.4	Understand subtraction as an unknown-addend problem. For example, subtract $10 - 8$ by finding the number that makes 10 when added to 8.	Mission 1, Topic G, Lessons 26-27 Mission 1, Topic H, Lessons 29, 30, 32 Mission 2, Topic B, Lessons 14-21 Mission 2, Topic C, Lesson 22
1.OA.5	Relate counting to addition and subtraction (e.g., by counting on 2 to add 2).	Mission 1, Topic A, Lessons 1-3 Mission 1, Topic B, Lessons 4-8

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 1, Topic D, Lessons 14-16 Mission 1, Topic G, Lessons 26-27 Mission 2, Topic A, Lessons 5, 9 Mission 2, Topic B, Lessons 12-19, 21
1.OA.6	Add and subtract within 20, demonstrating fluency with various strategies for addition and subtraction within 10. Strategies may include counting on; making ten, e.g., $8 + 6 = 8 + 2 + 4 = 10 + 4 = 14$; decomposing a number leading to a ten, e.g., $13 - 4 = 13 - 3 - 1 = 10 - 1 = 9$; using the relationship between addition and subtraction, e.g., knowing that $8 + 4 = 12$, one knows $12 - 8 = 4$; and creating equivalent but easier or known sums, e.g., adding $6 + 7$ by creating the known equivalent $6 + 6 + 1 = 12 + 1 = 13$.	Mission 1, Topic C, Lessons 9-13 Mission 1, Topic D, Lessons 14-16 Mission 1, Topic E, Lessons 17-20 Mission 1, Topic F, Lessons 21-24 Mission 1, Topic G, Lessons 26, 27 Mission 1, Topic I, Lessons 33-37 Mission 1, Topic J, Lessons 38, 39 Mission 2, Topic A, Lessons 1-11 Mission 2, Topic B, Lessons 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lessons 27-29
1.OA.7	Understand the meaning of the equal sign, and determine if equations involving addition and subtraction are true or false. For example, which of the following equations are true and which are false? $6 = 6$, $7 = 8 - 1$, $5 + 2 = 2 + 5$, $4 + 1 = 5 + 2$.	Mission 1, Topic E, Lessons 17-19 Mission 2, Topic A, Lessons 6, 10 Mission 2, Topic B, Lesson 20 Mission 2, Topic C, Lesson 25
1.OA.8	Determine the unknown whole number in an addition or subtraction equation relating three whole numbers. For example, determine the unknown number that makes the equation true in each of the equations $8 + ? = 11$, $5 = \diamond - 3$, $6 + 6 = \diamond$.	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lessons 29-32 Mission 2, Topic A, Lesson 10 Mission 2, Topic A, Lesson 20
Numbers & Operations in Base 10		
1.NBT.1	Count to 120, starting at any number less than 120. In this range, read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A, Lessons 1-6 Mission 6, Topic B, Lessons 4, 7, 9
1.NBT.2	Understand that the two digits of a two-digit number represent amounts of tens and ones. Understand the following as special cases: 10 can be thought of as a bundle of ten ones — called a “ten;” the numbers from 11 to 19 are composed of a ten and one, two, three, four, five, six, seven, eight, or nine ones; and the numbers 10, 20, 30, 40, 50, 60, 70, 80, 90 refer to one,	Mission 2, Topic D, Lessons 26-29 Mission 4, Topic A, Lessons 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lessons 3, 4, 8, 9

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	two, three, four, five, six, seven, eight, or nine tens (and 0 ones).	
1.NBT.3	Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	Mission 4, Topic B, Lessons 7-10 Mission 6, Topic B, Lesson 6
1.NBT.4	Add within 100, including adding a two-digit number and a one-digit number and adding a two-digit number and a multiple of 10, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; record the strategy with a written numerical method (drawings and, when appropriate, equations) and explain the reasoning used. Understand that when adding two-digit numbers, tens are added to tens; ones are added to ones; and sometimes it is necessary to compose a ten.	Mission 4, Topic C, Lessons 11, 12 Mission 4, Topic D, Lessons 13-18 Mission 4, Topic F, Lessons 23-29 Mission 6, Topic C, Lessons 10-17 Mission 6, Topic D, Lessons 18, 19
1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 4, Topic A, Lessons 5, 6 Mission 6, Topic B, Lesson 5
1.NBT.6	Subtract multiples of 10 in the range 10-90 from multiples of 10 in the range 10-90 (positive or zero differences), using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; relate the strategy to a written method and explain the reasoning used.	Mission 4, Topic C, Lesson 11 Mission 6, Topic C, Lesson 10
Measurement & Data		
1.MD.1	Order three objects by length; compare the lengths of two objects indirectly by using a third object.	Mission 3, Topic A, Lessons 1-3 Mission 3, Topic B, Lessons 5, 6 Mission 3, Topic C, Lesson 9
1.MD.2	Express the length of an object as a whole number of length units, by laying multiple copies of a shorter object (the length unit) end to end; understand that the length measurement of an object is the number of same-size length units that span it with no gaps or overlaps. Limit to contexts where the object being	Mission 3, Topic B, Lessons 4-6 Mission 3, Topic C, Lessons 7-9

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
	measured is spanned by a whole number of length units with no gaps or overlaps.	
1.MD.3	Work with time and money.	Mission 4, Topic A, Lesson 6 Mission 5, Topic D, Lessons 10-13 Mission 3, Topic C, Lesson 8 Mission 6, Topic E, Lessons 20-24
1.MD.4	Organize, represent, and interpret data with up to three categories; ask and answer questions about the total number of data points, how many in each category, and how many more or less are in one category than in another.	Mission 3, Topic D, Lessons 10-13
Geometry		
1.G.1	Distinguish between defining attributes, e.g., triangles are closed and three-sided, versus non-defining attributes, e.g., color, orientation, overall size; build and draw shapes that possess defining attributes.	Mission 5, Topic A, Lessons 1-3
1.G.2	Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) or three dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shape. Students do not need to learn formal names such as "right rectangular prism."	Mission 5, Topic B, Lessons 4-6
1.G.3	Partition circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of. Describe the whole as two of or four of the shares in real-world contexts. Understand for these examples that decomposing into more equal shares creates smaller shares.	Mission 5, Topic C, Lessons 7-9 Mission 5, Topic D, Lessons 10

2nd Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
2.OA.1	Use addition and subtraction within 100 to solve one- and two-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem.	Mission 1, Topic B, Lessons 3, 5, 6, 8 Mission 4, Topic A, Lessons 1-5 Mission 4, Topic B, Lessons 6-10 Mission 4, Topic C, Lessons 11-16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B Lessons 7, 8, 12, 13
2.OA.2	Fluently add and subtract within 20 using mental strategies. ² By end of Grade 2, know from memory all sums of two one-digit numbers.	Mission 1, Topic A, Lessons 1, 2 Mission 1, Topic B, Lessons 4, 7
2.OA.3	Determine whether a group of objects (up to 20) has an odd or even number of members, e.g., by pairing objects or counting them by 2s; write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D, Lessons 17-19
2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	Mission 6, Topic A, Lessons 1-4 Mission 6, Topic B, Lessons 5-9 Mission 6, Topic C, Lessons 10-15
Numbers & Operations in Base 10		
2.NBT.1	Understand that the three digits of a three-digit number represent amounts of hundreds, tens, and ones; e.g., 706 equals 7 hundreds, 0 tens, and 6 ones. Understand the following as special cases: a. 100 can be thought of as a bundle of ten tens - called a "hundred." b. The numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five,	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lessons 2, 3 Mission 3, Topic C, Lessons 4-7 Mission 3, Topic D, Lessons 8, 9 Mission 3, Topic E, Lessons 11-15

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	six, seven, eight, or nine hundreds (and 0 tens and 0 ones).	
2.NBT.2	Count forward and backward within 1,000 by ones, tens, and hundreds starting at any number; skip-count by 5s starting at any multiple of 5.	Mission 3, Topic B, Lessons 2, 3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lessons 8, 9 Mission 3, Topic E, Lessons 12-15 Mission 3, Topic G, Lessons 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NBT.3	Read and write numbers to 1,000 using base-ten numerals, number names, expanded form, and equivalent representations, e.g., 716 is $700 + 10 + 6$, or $6 + 700 + 10$, or 6 ones and 71 tens, etc.	Mission 3, Topic C, Lessons 4-7 Mission 3, Topic D, Lessons 8, 9 Mission 3, Topic E, Lessons 13, 14
2.NBT.4	Compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 3, Topic F, Lessons 16-18
2.NBT.5	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 1, Topic B, Lessons 3, 5, 6, 8 Mission 4, Topic A, Lessons 2, 3 Mission 4, Topic B, Lessons 6-8, 10 Mission 4, Topic C, Lessons 11-13 Mission 7, Topic B, Lessons 7, 8, 11-13
2.NBT.6	Add up to four two-digit numbers using strategies based on place value and properties of operations.	Mission 4, Topic D, Lesson 22
2.NBT.7	Add and subtract within 1,000, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction; record the strategy with a written numerical method (drawings and, when appropriate, equations) and explain the reasoning used. Understand that in adding or subtracting three-digit numbers, hundreds are added or subtracted from hundreds, tens are added or subtracted from tens, ones are added or subtracted from ones; and sometimes it is necessary to compose or decompose tens or hundreds.	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lessons 14, 15 Mission 4, Topic D, Lessons 17-21 Mission 4, Topic E, Lessons 23-28 Mission 4, Topic F, Lessons 29, 30 Mission 5, Topic A, Lessons 1-7 Mission 5, Topic B, Lessons 8-12 Mission 5, Topic C, Lessons 13-18 Mission 5, Topic D, Lessons 19, 20

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.NBT.8	Mentally add 10 or 100 to a given number 100–900, and mentally subtract 10 or 100 from a given number 100–900.	Mission 3, Topic G, Lessons 19, 20 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lessons 1, 2
2.NBT.9	Explain why addition and subtraction strategies work, using place value and the properties of operations. Explanations may be supported by drawings or objects.	Mission 4, Topic C, Lesson 11 Mission 4, Topic F, Lesson 30 Mission 5, Topic A, Lesson 7 Mission 5, Topic B, Lesson 12 Mission 5, Topic C, Lessons 13-19 Mission 5, Topic D, Lesson 20
Measurement & Data		
2.MD.1	Measure the length of an object by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A, Lessons 1-3 Mission 2, Topic B, Lessons 4, 5 Mission 2, Topic C, Lesson 6 Mission 7, Topic C, Lessons 14-17
2.MD.2	Measure the length of an object twice, using length units of different lengths for the two measurements; describe how the two measurements relate to the size of the unit chosen.	Mission 2, Topic C, Lesson 7 Mission 7, Topic D, Lesson 18
2.MD.3	Estimate lengths using units of inches, feet, centimeters, and meters.	Mission 2, Topic B, Lesson 5 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 17
2.MD.4	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard length unit.	Mission 2, Topic C, Lessons 6, 7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 19
2.MD.5	Use addition and subtraction within 100 to solve word problems involving lengths that are given in the same whole number units, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem. Drawings need not show details, but should show the mathematics in the problem.	Mission 2, Topic D, Lessons 8-10 Mission 7, Topic E, Lessons 20, 22
2.MD.6	Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points corresponding to the numbers 0, 1, 2, ..., and	Mission 2, Topic D, Lesson 8 Mission 7, Topic A, Lesson 3 Mission 7, Topic E, Lessons 21, 22 Mission 7, Topic F, Lessons 24

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	represent whole-number sums and differences within 100 on a number line diagram.	
2.MD.7	Tell and write time from analog and digital clocks to the nearest five minutes, using a.m. and p.m.	Mission 8, Topic D, Lessons 13-15
2.MD.8	Solve problems with money.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-13
2.MD.9	Generate measurement data by measuring lengths of several objects to the nearest whole unit or by making repeated measurements of the same object. Show the measurements by creating a line plot, where the horizontal scale is marked off in whole number units.	Mission 7, Topic F, Lessons 23-26
2.MD.10	Organize, represent, and interpret data with up to four categories; complete picture graphs when single-unit scales are provided; complete bar graphs when single-unit scales are provided; solve simple put-together, take-apart, and compare problems in a graph.	Mission 7, Topic A, Lessons 1-5
Geometry		
2.G.1	Recognize and identify triangles, quadrilaterals, pentagons, and hexagons based on the number of sides or vertices. Recognize and identify cubes, rectangular prisms, cones, and cylinders.	Mission 8, Topic A, Lessons 1-5 Mission 8, Topic B, Lesson 6
2.G.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of them.	Mission 6, Topic C, Lessons 14, 15
2.G.3	Partition circles and rectangles into two, three, or four equal shares; describe the shares using the words halves, thirds, or fourths and quarters, and use the phrases half of, third of, or fourth of and quarter of. Describe the whole as two halves, three thirds, or four fourths in real-world contexts. Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B, Lessons 7, 8 Mission 8, Topic C, Lessons 9-12 Mission 8, Topic D, Lesson 13

3rd Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
3.OA.1	Interpret products of whole numbers, e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each. For example, describe a context in which a total number of objects can be expressed as 5×7 .	Mission 1, Topic A, Lessons 1-3 Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lessons 7-10 Mission 1, Topic E, Lessons 15, 17 Mission 3, Topic D, Lesson 12 Mission 3, Topic E, Lesson 16
3.OA.2	Interpret whole-number quotients of whole numbers, e.g., interpret $56 \div 8$ as the number of objects in each share when 56 objects are partitioned equally into 8 shares, or as a number of shares when 56 objects are partitioned into equal shares of 8 objects each. For example, describe a context in which a number of shares or a number of groups can be expressed as $56 \div 8$.	Mission 1, Topic B, Lessons 4-6 Mission 1, Topic D, Lessons 12, 13 Mission 1, Topic E, Lesson 17
3.OA.3	Use multiplication and division within 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem. Drawings need not show details, but should show the mathematics in the problem.	Mission 1, Topic B, Lessons 4-6 Mission 1, Topic C, Lessons 8, 9 Mission 1, Topic D, Lessons 11-13 Mission 1, Topic E, Lessons 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lessons 2, 3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 15
3.OA.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers. For example, determine the unknown number that makes the equation true in each of the equations $8 \times ? = 48$, $5 = ? \div 3$, $6 \times 6 = ?$.	Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 8 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 17 Mission 3, Topic A, Lesson 3 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
3.OA.5	Apply properties of operations as strategies to multiply and divide.2 Examples: If $6 \times 4 = 24$ is known, then $4 \times 6 = 24$ is also known. (Commutative property of multiplication.) $3 \times 5 \times 2$ can be found by $3 \times 5 = 15$,	Mission 1, Topic C, Lessons 7-10 Mission 1, Topic E, Lessons 15, 16 Mission 1, Topic F, Lessons 18, 19 Mission 3, Topic A, Lessons 1, 2

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	then $15 \times 2 = 30$, or by $5 \times 2 = 10$, then $3 \times 10 = 30$. (Associative property of multiplication.) Knowing that $8 \times 5 = 40$ and $8 \times 2 = 16$, one can find 8×7 as $8 \times (5 + 2) = (8 \times 5) + (8 \times 2) = 40 + 16 = 56$. (Distributive property.) Students need not use formal terms for these properties.	Mission 3, Topic B, Lessons 4-6 Mission 3, Topic C, Lessons 9, 10 Mission 3, Topic D, Lessons 12-14 Mission 3, Topic E, Lessons 16-17 Mission 3, Topic F, Lesson 20
3.OA.6	Understand division as an unknown-factor problem. For example, find $32 \div 8$ by finding the number that makes 32 when multiplied by 8.	Mission 1, Topic B, Lesson 6 Mission 1, Topic D, Lesson 11 Mission 1, Topic E, Lesson 17 Mission 3, Topic D, Lessons 12, 13 Mission 3, Topic E, Lesson 16
3.OA.7	Fluently multiply and divide within 100, using strategies such as the relationship between multiplication and division, e.g., knowing that $8 \times 5 = 40$, one knows $40 \div 5 = 8$, or properties of operations. Limit to division without remainders. By the end of Grade 3, know from memory all products of two one-digit numbers.	Mission 1, Topic D, Lessons 11-13 Mission 1, Topic E, Lessons 14, 16 Mission 3, Topic B, Lessons 4, 5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lessons 12, 13 Mission 3, Topic E, Lessons 16, 17 Mission 3, Topic F, Lesson 19
3.OA.8	Solve two-step word problems using the four operations. Represent these problems using equations with a letter or a symbol, which stands for the unknown quantity. Assess the reasonableness of answers using mental computation and estimation strategies including rounding. This standard is limited to problems posed with whole numbers and having whole number answers. Students may use parentheses for clarification since algebraic order of operations is not expected.	Mission 1, Topic F, Lessons 20, 21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lessons 1-3
3.OA.9	Identify arithmetic patterns (including patterns in the addition table or multiplication table), and explain them using properties of operations. For example, observe that 4 times a number is always even, and explain why 4 times a number can be decomposed into two equal addends.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lessons 13, 14 Mission 3, Topic E, Lessons 16, 17 Mission 3, Topic F, Lesson 19
Numbers & Operations in Base 10		
3.NBT.1	Use place value understanding to round whole numbers to the nearest 10 or 100.	Mission 2, Topic C, Lessons 12-14 Mission 2, Topic D, Lesson 17 Mission 2, Topic E, Lessons 20, 21

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.NBT.2	Fluently add and subtract within 1000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A, Lessons 4, 5 Mission 2, Topic B, Lessons 10, 11 Mission 2, Topic D, Lessons 15-17 Mission 2, Topic E, Lessons 18-21
3.NBT.3	Multiply one-digit whole numbers by multiples of 10 in the range 10–90 (e.g., 9×80 , 5×60) using strategies based on place value and properties of operations.	Mission 3, Topic F, Lessons 19-20
3.NF.1	Understand a fraction $\frac{1}{b}$ as the quantity formed by 1 part when a whole is partitioned into b equal parts; understand a fraction $\frac{a}{b}$ as the quantity formed by a parts of size $\frac{1}{b}$.	Mission 5, Topic B, Lessons 5-9 Mission 5, Topic C, Lessons 10-13 Mission 5, Topic E, Lessons 20, 22, 24, 25, 27 Mission 5, Topic F, Lessons 28, 29
3.NF.2	Understand a fraction as a number on the number line; represent fractions on a number line diagram.	Mission 5, Topic D, Lessons 14-17, 19 Mission 5, Topic E, Lessons 21, 23, 26
3.NF.3	Explain equivalence of fractions in special cases, and compare fractions by reasoning about their size.	Mission 5, Topic C, Lessons 10-11, 13 Mission 5, Topic D, Lessons 16-19 Mission 5, Topic E, Lessons 20-27 Mission 5, Topic F, Lessons 28-29
Measurement & Data		
3.MD.1	Work with time and money.	Mission 2, Topic A, Lessons 1-5 Mission 2, Topic D, Lessons 15-17 Mission 2, Topic E, Lessons 18, 21
3.MD.2	Measure and estimate liquid volumes and masses of objects using standard units of grams, kilograms, and liters. Add, subtract, multiply, or divide whole numbers to solve one-step word problems involving masses or volumes that are given in the same units, e.g., by using drawings (such as a beaker with a measurement scale) to represent the problem.	Mission 3, Topic B, Lesson 6 Mission 2, Topic B, Lessons 7-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lessons 15-17 Mission 2, Topic E, Lesson 18-21
3.MD.3	Create scaled picture graphs to represent a data set with several categories. Create scaled bar graphs to represent a data set with several categories. Solve	Mission 6, Topic A, Lessons 1-4

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3rd Grade Standards		Lessons
	two-step “how many more” and “how many less” problems using information presented in the scaled graphs. For example, create a bar graph in which each square in the bar graph might represent 5 pets, then determine how many more/less in two given categories.	
3.MD.4	Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by creating a line plot, where the horizontal scale is marked off in appropriate units—whole numbers, halves, or quarters.	Mission 6, Topic B, Lessons 5-9 Mission 7, Topic D, Lessons 19, 22 Mission 7, Topic E, Lesson 26
3.MD.5	Recognize area as an attribute of plane figures and understand concepts of area measurement.	Mission 4, Topic A, Lessons 1-4 Mission 4, Topic B, Lessons 5-7 Mission 4, Topic D, Lesson 13
3.MD.6	Measure areas by counting unit squares (square cm, square m, square in, square ft, and improvised units).	Mission 4, Topic A, Lessons 1-4 Mission 4, Topic B, Lessons 5-7 Mission 4, Topic D, Lesson 13
3.MD.7	Relate area to the operations of multiplication and addition.	Mission 4, Topic A, Lesson 4 Mission 4, Topic B, Lessons 5-8 Mission 4, Topic C, Lessons 9-11 Mission 4, Topic D, Lessons 12-16 Mission 7, Topic E, Lessons 28-30
3.MD.8	Solve real world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths, finding an unknown side length, and exhibiting rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 7, Topic C, Lessons 10-17 Mission 7, Topic D, Lessons 18-22 Mission 7, Topic E, Lessons 23-30
Geometry		
3.G.1	Draw and describe triangles, quadrilaterals (rhombuses, rectangles, and squares), and polygons (up to 8 sides) based on the number of sides and the presence or absence of square corners (right angles).	Mission 7, Topic B, Lessons 4-9
3.G.2	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole.	Mission 5, Topic A, Lessons 1-4 Mission 5, Topic B, Lessons 5-9

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3rd Grade Standards	Lessons
For example, partition a shape into 4 parts with equal area, and describe the area of each part as $\frac{1}{4}$ of the area of the shape.	Mission 5, Topic C, Lessons 10-13 Mission 5, Topic E, Lessons 24-27 Mission 5, Topic F, Lessons 28, 29

4th Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
4.OA.1	Interpret a multiplication equation as a comparison, e.g., interpret $35 = 5 \times 7$ as a statement that 35 is 5 times as many as 7 and 7 times as many as 5. Represent verbal statements of multiplicative comparisons as multiplication equations.	Mission 1, Topic A, Lesson 1
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison, e.g., by using drawings and equations with a symbol for the unknown number to represent the problem, distinguishing multiplicative comparison from additive comparison. Drawings need not show details, but should show the mathematics in the problem.	Mission 3, Topic A, Lessons 2, 3 Mission 3, Topic B, Lessons 5, 6 Mission 3, Topic C, Lessons 9-11 Mission 3, Topic D, Lessons 12, 13 Mission 3, Topic G, Lessons 26 Mission 5, Topic G, Lessons 39, 40 Mission 7, Topic A, Lessons 4 Mission 7, Topic C, Lessons 14
4.OA.3	Solve multistep word problems posed with whole numbers and having whole-number answers using the four operations, including problems in which remainders must be interpreted. Represent these problems using equations with a letter standing for the unknown quantity. Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 3, Topic D, Lessons 12, 13 Mission 3, Topic E Lessons 14, 19 Mission 3, Topic G, Lessons 26 Mission 3, Topic G, Lessons 31, 32
4.OA.4	Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.	Mission 3, Topic F, Lessons 22-24 Mission 5, Topic H, Lesson 41
Numbers & Operations in Base 10		
4.NBT.1	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right by applying concepts of place value, multiplication, or division.	Mission 1, Topic A, Lessons 2, 3

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.NBT.2	Read and write multi-digit whole numbers using standard form, word form, and expanded form. Compare two multi-digit numbers based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons. Grade 4 expectations in this domain are limited to whole numbers less than or equal to 1,000,000.	Mission 1, Topic A, Lesson 3 Mission 1, Topic B, Lessons 4-6
4.NBT.3	Use place value understanding to round multi-digit whole numbers to any place through 1,000,000.	Mission 1, Topic C, Lessons 7-10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lessons 16, 17 Mission 1, Topic F, Lesson 19
4.NBT.4	Fluently add and subtract multi-digit whole numbers using a standard algorithm.	Mission 1, Topic D, Lessons 11-13 Mission 1, Topic E, Lessons 14-17 Mission 1, Topic F, Lessons 18, 19
4.NBT.5	Multiply a whole number of up to four digits by a one-digit whole number, and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B, Lessons 4-6 Mission 3, Topic C, Lessons 7-11 Mission 3, Topic H, Lessons 34-38
4.NBT.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic E, Lessons 14-21 Mission 3, Topic G, Lessons 26-33
Numbers & Operations - Fractions		
4.NF.1	Explain why a fraction $\frac{a}{b}$ is equivalent to a fraction $\frac{n \times a}{n \times b}$ by using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	Mission 5, Topic B, Lessons 7-11 Mission 5, Topic C, Lessons 14, 15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12
4.NF.2	Compare two fractions with different numerators and different denominators, e.g., by creating common	Mission 5, Topic C, Lessons 12-15 Mission 5, Topic E, Lessons 26, 27

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	denominators or numerators, or by comparing to a benchmark fraction such as $\frac{1}{2}$. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions, e.g., by using a visual fraction model.	
4.NF.3	Understand a fraction $\frac{a}{b}$ with $a > 1$ as a sum of fractions $\frac{1}{b}$.	Mission 5, Topic A, Lessons 1-6 Mission 5, Topic D, Lessons 16-21 Mission 5, Topic E, Lessons 22-25 Mission 5, Topic F, Lesson 29-34
4.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number.	Mission 5, Topic A, Lessons 3, 5, 6 Mission 5, Topic E, Lessons 23-25 Mission 5, Topic G, Lessons 35-40
4.NF.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100. For example, express $\frac{3}{10}$ as $\frac{30}{100}$, and add $\frac{3}{10} + \frac{4}{100} = \frac{34}{100}$. In general, students who can generate equivalent fractions can develop strategies for adding fractions with unlike denominators, but addition and subtraction with unlike denominators is not a requirement at this grade.	Mission 6, Topic B, Lesson 4 Mission 6, Topic D, Lessons 12-14
4.NF.6	Use decimal notation for fractions with denominators 10 or 100. For example, rewrite 0.62 as $\frac{62}{100}$; describe a length as 0.62 meters; locate 0.62 on a number line diagram.	Mission 6, Topic A, Lessons 1-3 Mission 6, Topic B, Lessons 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lessons 12-14 Mission 6, Topic E, Lesson 15
4.NF.7	Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions, e.g., by using a visual model.	Mission 6, Topic C, Lessons 9-11
Measurement & Data		

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.MD.1	Know relative sizes of the metric measurement units within one system of units. Metric units include kilometer, meter, centimeter, and millimeter; kilogram and gram; and liter and milliliter. Express a larger measurement unit in terms of a smaller unit. Record measurement conversions in a two-column table. For example, express the length of a 4-meter rope in centimeters. Because 1 meter is 100 times as long as a 1 centimeter, a two-column table of meters and centimeters includes the number pairs 1 and 100, 2 and 200, 3 and 300,...	Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lessons 1-3 Mission 7, Topic B, Lessons 6-9 Mission 7, Topic C, Lessons 12, 13
4.MD.2	Solve real-world problems involving money, time, and metric measurement.	Mission 5, Topic G, Lesson 40 Mission 6, Topic D, Lesson 14 Mission 6, Topic E, Lessons 15, 16 Mission 7, Topic A, Lessons 1-5 Mission 7, Topic B, Lessons 6-11 Mission 7, Topic C, Lesson 14
4.MD.3	Develop efficient strategies to determine the area and perimeter of rectangles in real-world situations and mathematical problems. For example, given the total area and one side length of a rectangle, solve for the unknown factor, and given two adjacent side lengths of a rectangle, find the perimeter.	Mission 3, Topic A, Lessons 1-3 Mission 7, Topic D, Lessons 15, 16
4.MD.4	Display and interpret data in graphs (picture graphs, bar graphs, and line plots) to solve problems using numbers and operations for this grade.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40
4.MD.5	Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint, and understand concepts of angle measurement.	Mission 4, Topic A, Lesson 1 Mission 4, Topic B, Lessons 5-7
4.MD.6	Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.	Mission 4, Topic B, Lessons 5-8
4.MD.7	Recognize angle measure as additive. When an angle is decomposed into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. Solve addition and subtraction problems to find unknown angles on a diagram in real world and mathematical problems, e.g., by using an	Mission 4, Topic C, Lessons 9-11

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
equation with a symbol for the unknown angle measure.		
Geometry		
4.G.1	Draw points, lines, line segments, rays, angles (right, acute, and obtuse), and perpendicular and parallel lines. Identify these in two-dimensional figures.	Mission 4, Topic A, Lessons 1-4 Mission 4, Topic D, Lessons 13-15
4.G.2	Classify two-dimensional figures based on the presence or absence of parallel or perpendicular lines or the presence or absence of angles of a specified size.	Mission 4, Topic D, Lesson 13-15

5th Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
5.OA.1	Use parentheses in numerical expressions, and evaluate expressions with this symbol. Formal use of algebraic order of operations is not necessary.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lessons 3-5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.2	Write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them. For example, express the calculation “add 8 and 7, then multiply by 2” as $2 \times (8 + 7)$. Recognize that $3 \times (18,932 + 921)$ is three times as large as $18,932 + 921$, without having to calculate the indicated sum or product.	Mission 2, Topic B, Lessons 3, 4 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.3	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane. For example, given the rule “Add 3” and the starting number 0, and given the rule “Add 6” and the starting number 0, generate terms in the resulting sequences, and observe that the terms in one sequence are twice the corresponding terms in the other sequence. Explain informally why this is so.	Mission 6, Topic B, Lesson 7
Numbers & Operations in Base 10		
5.NBT.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and $1/10$ of what it represents in the place to its left.	Mission 1, Topic A, Lessons 1-4 Mission 2, Topic A, Lesson 2 Mission 2, Topic E, Lesson 16
5.NBT.2	Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of	Mission 1, Topic A, Lessons 1-4 Mission 2, Topic A, Lesson 1 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lessons 13, 14 Mission 2, Topic G, Lesson 24

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	10. Use whole-number exponents to denote powers of 10.	
5.NBT.3	Read, write, and compare decimals to thousandths.	Mission 1, Topic B, Lessons 5, 6 Mission 1, Topic D, Lessons 9, 10 Mission 1, Topic E, Lessons 11, 12 Mission 1, Topic F, Lessons 13-15
5.NBT.4	Use place value understanding to round decimals to any place, millions through hundredths.	Mission 1, Topic C, Lessons 7, 8
5.NBT.5	Fluently multiply multi-digit whole numbers using a standard algorithm.	Mission 2, Topic B, Lessons 5-9 Mission 2, Topic C, Lessons 10-12 Mission 2, Topic F, Lessons 20-23
5.NBT.6	Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 2, Topic E, Lessons 16-18 Mission 2, Topic F, Lessons 19-23 Mission 2, Topic H, Lessons 28, 29 Mission 2, Topic G, Lessons 30, 31
5.NBT.7	Solve real-world problems by adding, subtracting, multiplying, and dividing decimals using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction, or multiplication and division; relate the strategy to a written method and explain the reasoning used.	Mission 1, Topic D, Lessons 9, 10 Mission 1, Topic E, Lessons 11, 12 Mission 1, Topic F, Lessons 13-16 Mission 2, Topic C, Lessons 10-12 Mission 2, Topic D, Lessons 13, 14 Mission 2, Topic G, Lessons 24-27 Mission 2, Topic H, Lessons 28, 29 Mission 4, Topic E, Lessons 17, 18 Mission 4, Topic G, Lessons 29-31
Numbers & Operations - Fractions		
5.NF.1	Add and subtract fractions with unlike denominators (including mixed numbers and fractions greater than 1) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators. For example, use visual models and properties of operations to show $\frac{2}{3} + \frac{5}{4} = \frac{8}{12} + \frac{15}{12} = \frac{23}{12}$. In	Mission 3, Topic A, Lessons 1, 2 Mission 3, Topic B, Lessons 3-6 Mission 3, Topic C, Lessons 8-12 Mission 3, Topic D, Lesson 14

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	general, $a/b + c/d = (a/b \times d/d) + (c/d \times b/b) = (ad + bc)/bd$.	
5.NF.2	Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators, e.g., by using visual fraction models or equations to represent the problem. Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers. For example, recognize an incorrect result $2/5 + 1/2 = 3/7$, by observing that $3/7 < 1/2$.	Mission 3, Topic B, Lessons 3-7 Mission 3, Topic C, Lessons 8-12 Mission 3, Topic D, Lessons 13-15 Mission 4, Topic D, Lessons 10-12
5.NF.3	Interpret a fraction as division of the numerator by the denominator ($a/b = a \div b$). Solve word problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers, e.g., by using visual fraction models or equations to represent the problem. For example, interpret $3/4$ as the result of dividing 3 by 4, noting that $3/4$ multiplied by 4 equals 3, and that when 3 wholes are shared equally among 4 people each person has a share of size $3/4$. If 9 people want to share a 50-pound sack of rice equally by weight, how many pounds of rice should each person get? Between what two whole numbers does your answer lie?	Mission 4, Topic B, Lessons 2-5
5.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction.	Mission 4, Topic C, Lessons 6-9 Mission 4, Topic E, Lessons 13-15, 17, 18 Mission 5, Topic C, Lessons 10-15
5.NF.5	Interpret multiplication as scaling (resizing.)	Mission 4, Topic F, Lessons 21-23 Mission 4, Topic H, Lesson 32
5.NF.6	Solve real world problems involving multiplication of fractions and mixed numbers, e.g., by using visual fraction models or equations to represent the problem.	Mission 4, Topic C, Lessons 6-7 Mission 4, Topic D, Lessons 10-12 Mission 4, Topic E, Lessons 13-18 Mission 4, Topic F, Lessons 22-24 Mission 5, Topic C, Lessons 13-15
5.NF.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions. Students able to multiply	Mission 4, Topic G, Lessons 25-33

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
fractions in general can develop strategies to divide fractions in general, by reasoning about the relationship between multiplication and division, but division of a fraction by a fraction is not a requirement at this grade.		
Measurement & Data		
5.MD.1	Know relative sizes of these U.S. customary measurement units: pounds, ounces, miles, yards, feet, inches, gallons, quarts, pints, cups, fluid ounces, hours, minutes, and seconds. Convert between pounds and ounces; miles and feet; yards, feet, and inches; gallons, quarts, pints, cups, and fluid ounces; hours, minutes, and seconds in solving multi-step, real-world problems.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lessons 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lessons 19, 20 Mission 5, Topic B, Lesson 5
5.MD.2	Display and interpret data in graphs (picture graphs, bar graphs, and line plots) to solve problems using numbers and operations for this grade, e.g., including U.S. customary units in fractions $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, or decimals.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10
5.MD.3	Recognize volume as an attribute of solid figures and understand concepts of volume measurement.	Mission 5, Topic A, Lessons 1-3 Mission 5, Topic B, Lesson 5
5.MD.4	Measure volumes by counting unit cubes, using cubic cm, cubic in, cubic ft, and improvised units.	Mission 5, Topic A, Lessons 1-3
5.MD.5	Relate volume to the operations of multiplication and addition and solve real-world and mathematical problems involving volume.	Mission 5, Topic B, Lessons 4-9
Geometry		
5.G.1	Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the 0 on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. Understand that the first number indicates how far to travel from the origin in the	Mission 6, Topic A, Lessons 1-6 Mission 6, Topic B, Lessons 7-12 Mission 6, Topic C, Lessons 13-17 Mission 6, Topic D, Lessons 18, 20

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	direction of one axis, and the second number indicates how far to travel in the direction of the second axis, with the convention that the names of the two axes and the coordinates correspond, e.g., x -axis and x -coordinate, y -axis and y -coordinate.	
5.G.2	Represent real world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C, Lessons 13-17 Mission 6, Topic D, Lessons 18-20
5.G.3	Identify and describe commonalities and differences between types of triangles based on angle measures (equiangular, right, acute, and obtuse triangles) and side lengths (isosceles, equilateral, and scalene triangles).	Mission 4, Topic D, Lesson 13
5.G.4	Identify and describe commonalities and differences between types of quadrilaterals based on angle measures, side lengths, and the presence or absence of parallel and perpendicular lines, e.g., squares, rectangles, parallelograms, trapezoids, and rhombuses.	Mission 5, Topic D, Lessons 16-21

6th Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities. For example, "The ratio of wings to beaks in the bird house at the zoo was 2:1, because for every 2 wings there was 1 beak." "For every vote candidate A received, candidate C received nearly three votes."	Mission 2, Topic A, Lesson 1, 2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.RP.2	Understand the concept of a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$, and use rate language in the context of a ratio relationship. For example, "This recipe has a ratio of 3 cups of flour to 4 cups of sugar, so there is $3/4$ cup of flour for each cup of sugar." "We paid \$75 for 15 hamburgers, which is a rate of \$5 per hamburger."	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
6.RP.3	Use ratio and rate reasoning to solve real-world and mathematical problems, e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations.	Mission 2, Topic C, Lessons 6-10 Mission 2, Topic D, Lessons 11-14 Mission 2, Topic E, Lessons 15, 16 Mission 2, Topic F, Lesson 17 Mission 3, Topic B, Lessons 3, 4 Mission 3, Topic C, Lessons 5-9 Mission 3, Topic D, Lessons 10-16 Mission 3, Topic E, Lesson 17 Mission 9, Topic A, Lessons 4-6
The Number System		
6.NS.1	Interpret and compute quotients of fractions, and solve word problems involving division of fractions by fractions, e.g., by using visual fraction models and equations to represent the problem. For example, create a story context for $(2/3) \div (3/4)$ and use a visual fraction model to show the quotient; use the relationship between multiplication and division to explain that $(2/3) \div (3/4) = 8/9$ because $3/4$ of $8/9$ is $2/3$. (In general, $(a/b) \div (c/d) = ad/bc$.) How much chocolate	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lessons 4-9 Mission 4, Topic C, Lessons 10, 11 Mission 4, Topic D, Lessons 12-14 Mission 4, Topic E, Lessons 16, 17

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	will each person get if 3 people share $\frac{1}{2}$ lb of chocolate equally? How many $\frac{3}{4}$ -cup servings are in $\frac{2}{3}$ of a cup of yogurt? How wide is a rectangular strip of land with length $\frac{3}{4}$ mi and area $\frac{1}{2}$ square mi?	
6.NS.2	Fluently divide multi-digit numbers using a standard algorithm.	Mission 5, Topic D, Lessons 9-11
6.NS.3	Fluently add, subtract, multiply, and divide multi-digit decimals using a standard algorithm for each operation.	Mission 5, Topic B, Lessons 2-4 Mission 5, Topic C, Lessons 7, 8 Mission 5, Topic D, Lessons 12, 13 Mission 5, Topic E, Lessons 14, 15 Mission 6, Topic A, Lesson 4 Mission 8, Topic C, Lesson 12 Mission 9, Topic A, Lesson 6
6.NS.4	Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers with no common factor. For example, express $36 + 8$ as $4(9 + 2)$.	Mission 6, Topic D, Lessons 16-18
6.NS.5	Understand that positive and negative numbers are used together to describe quantities having opposite directions or values, e.g., temperature above/below zero, elevation above/below sea level, credits/debits, positive/negative electric charge; use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of 0 in each situation.	Mission 7, Topic A, Lessons 1, 5
6.NS.6	Understand a rational number as a point on the number line. Extend number line diagrams and coordinate axes familiar from previous grades to represent points on the line and in the plane with negative number coordinates.	Mission 7, Topic A, Lessons 1-2, 4, 7 Mission 7, Topic C, Lesson 11-15
6.NS.7	Understand ordering and absolute value of rational numbers.	Mission 7, Topic A, Lessons 3, 4, 6, 7 Mission 7, Topic B, Lessons 8, 9 Mission 7, Topic C, Lesson 13

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.NS.8	Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C, Lessons 11, 13-15
Expressions & Equations		
6.EE.1	Write and evaluate numerical expressions involving whole-number exponents.	Mission 1, Topic F, Lessons 17-18 Mission 6, Topic C, Lessons 12-15
6.EE.2	Write, read, and evaluate expressions in which letters stand for numbers.	Mission 6, Topic B, Lessons 6, 10, 11 Mission 1, Topic B, Lesson 5 Mission 1, Topic C, Lesson 9 Mission 1, Topic F, Lesson 18 Mission 6, Topic C, Lessons 14, 15
6.EE.3	Apply the properties of operations to generate equivalent expressions. For example, apply the distributive property to the expression $3(2 + x)$ to produce the equivalent expression $6 + 3x$; apply the distributive property to the expression $24x + 18y$ to produce the equivalent expression $6(4x + 3y)$; apply properties of operations to $y + y + y$ to produce the equivalent expression $3y$.	Mission 6, Topic B, Lessons 10, 11
6.EE.4	Identify when two expressions are equivalent, i.e., when the two expressions name the same number regardless of which value is substituted into them. For example, the expressions $y + y + y$ and $3y$ are equivalent because they name the same number regardless of which number y stands for.	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lessons 8, 10, 11
6.EE.5	Understand solving an equation or inequality as a process of answering a question: which values from a specified set, if any, make the equation or inequality true? Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A, Lessons 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lessons 9-10
6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or	Mission 6, Topic A, Lessons 1, 3-5 Mission 6, Topic B, Lessons 6, 7

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	mathematical problem; understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	Mission 7, Topic B, Lessons 8, 10
6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form $x + p = q$ and $px = q$ for cases in which p , q , and x are all nonnegative rational numbers.	Mission 6, Topic A, Lessons 3-5 Mission 6, Topic B, Lesson 7
6.EE.8	Write an inequality of the form $x > c$ or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of the form $x > c$ or $x < c$ have infinitely many solutions; represent solutions of such inequalities on number line diagrams.	Mission 7, Topic B, Lessons 9, 10
6.EE.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another; write an equation to express one quantity, thought of as the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation. For example, in a problem involving motion at constant speed, list and graph ordered pairs of distances and times, and write the equation $d = 65t$ to represent the relationship between distance and time.	Mission 6, Topic D, Lessons 16-18
Geometry		
6.G.1	Through composition into rectangles or decomposition into triangles, find the area of right triangles, other triangles, special quadrilaterals, and polygons; apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic A, Lessons 2-3 Mission 1, Topic B, Lessons 4-6 Mission 1, Topic C, Lessons 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
6.G.2	Find the volume of a right rectangular prism with fractional edge lengths by packing it with unit cubes of the appropriate unit fraction edge lengths, and show that the volume is the same as would be found by multiplying the edge lengths of the prism. Apply the formulas $V = lwh$ and $V = Bh$ to find volumes of right	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lessons 14, 15 Mission 4, Topic E, Lesson 17

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems.	
6.G.3	Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side joining points with the same first coordinate or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 7, Topic C, Lesson 15
6.G.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lessons 12-16 Mission 1, Topic F, Lesson 18 Mission 1, Topic G, Lesson 19
Statistics & Probability		
6.SP.1	Develop statistical reasoning by using the GAISE model: a. Formulate Questions: Recognize and formulate a statistical question as one that anticipates variability and can be answered with quantitative data. For example, "How old am I?" is not a statistical question, but "How old are the students in my school?" is a statistical question because of the variability in students' ages. (GAISE Model, step 1) b. Collect Data: Design and use a plan to collect appropriate data to answer a statistical question. (GAISE Model, step 2) c. Analyze Data: Select appropriate graphical methods and numerical measures to analyze data by displaying variability within a group, comparing individual to individual, and comparing individual to group. (GAISE Model, step 3) d. Interpret Results: Draw logical conclusions from the data based on the original question. (GAISE Model, step 4)	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lessons 3, 6, 7 Mission 8, Topic D, Lesson 17
6.SP.2	Understand that a set of data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.	Mission 8, Topic B, Lessons 4, 5, 7, 8 Mission 8, Topic C, Lesson 11 Mission 8, Topic E, Lesson 18

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.SP.3	Recognize that a measure of center for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary with a single number.	Mission 8, Topic B, Lesson 6 Mission 8, Topic C, Lessons 9-11
6.SP.4	Display numerical data in plots on a number line, including dot plots (line plots), histograms, and box plots.	Mission 8, Topic B, Lessons 3-8 Mission 8, Topic D, Lessons 16, 17
6.SP.5	Summarize numerical data sets in relation to their context.	Mission 8, Topic D, Lesson 17 Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lessons 3-7 Mission 8, Topic C, Lessons 9-12 Mission 8, Topic D, Lessons 13-16 Mission 8, Topic E, Lesson 18

7th Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.1	Compute unit rates associated with ratios of fractions, including ratios of lengths, areas and other quantities measured in like or different units. For example, if a person walks $\frac{1}{2}$ mile in each $\frac{1}{4}$ hour, compute the unit rate as the complex fraction $\frac{1/2}{1/4}$ miles per hour, equivalently 2 miles per hour.	Mission 2, Topic C, Lesson 8 Mission 4, Topic A, Lessons 2, 3 Mission 9, Topic B, Lesson 5
7.RP.2	Recognize and represent proportional relationships between quantities.	Mission 2, Topic A, Lessons 2-5 Mission 2, Topic B, Lessons 4-6 Mission 2, Topic C, Lessons 7-9 Mission 2, Topic D, Lessons 10-13 Mission 2, Topic E, Lessons 14, 15 Mission 3, Topic A, Lesson 3, 5 Mission 4, Topic A, Lessons 3-5 Mission 5, Topic C, Lessons 9, 12 Mission 5, Topic D, Lesson 14 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 5
7.RP.3	Use proportional relationships to solve multistep ratio and percent problems. Examples: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.	Mission 3, Topic A, Lesson 5 Mission 4, Topic B, Lessons 6-9 Mission 4, Topic C, Lessons 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lessons 1-4 Mission 9, Topic B, Lessons 6, 8 Mission 9, Topic C, Lesson 13
The Number System		
7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lessons 2-7 Mission 6, Topic D, Lesson 18 Mission 7, Topic B, Lesson 6
7.NS.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers.	Mission 4, Topic A, Lesson 5 Mission 5, Topic A, Lessons 1, 5 Mission 5, Topic C, Lessons 9-11

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
		Mission 8, Topic D, Lesson 16 Mission 9, Topic A, Lesson 4
7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers. Computations with rational numbers extend the rules for manipulating fractions to complex fractions.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 12 Mission 5, Topic D, Lessons 13, 14 Mission 5, Topic E, Lessons 15, 16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
Expression & Equations		
7.EE.1	Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.	Mission 6, Topic D, Lessons 18-22 Mission 9, Topic B, Lesson 7
7.EE.2	In a problem context, understand that rewriting an expression in an equivalent form can reveal and explain properties of the quantities represented by the expression and can reveal how those quantities are related. For example, a discount of 15% (represented by $p - 0.15p$) is equivalent to $(1 - 0.15)p$, which is equivalent to $0.85p$ or finding 85% of the original price.	Mission 6, Topic B, Lesson 12
7.EE.4	Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities.	Mission 5, Topic C, Lesson 15 Mission 5, Topic E, Lessons 15, 16 Mission 6, Topic A, Lessons 4-5 Mission 6, Topic B, Lessons 7-12 Mission 6, Topic C, Lessons 14-17 Mission 7, Topic A, Lesson 5 Mission 9, Topic A, Lesson 3
Geometry		
7.G.1	Solve problems involving similar figures with right triangles, other triangles, and special quadrilaterals.	Mission 1, Topic A, Lessons 1-6 Mission 1, Topic B, Lessons 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
		Mission 9, Topic C, Lesson 13
7.G.2	Draw (freehand, with ruler and protractor, and with technology) geometric figures with given conditions.	Mission 3, Topic A, Lesson 2 Mission 7, Topic B, Lessons 6-10 Mission 7, Topic D, Lesson 17
7.G.3	Describe the two-dimensional figures that result from slicing three-dimensional figures, as in plane sections of right rectangular prisms and right rectangular pyramids.	Mission 7, Topic C, Lessons 11, 13
7.G.4	Work with circles.	Mission 3, Topic A, Lessons 3-5 Mission 3, Topic B, Lessons 7-9 Mission 3, Topic C, Lessons 10, 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lessons 11, 12
7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	Mission 7, Topic A, Lessons 2-5
7.G.6	Solve real-world and mathematical problems involving area, volume, and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lessons 12-16 Mission 7, Topic D, Lesson 17 Mission 9, Topic A, Lessons 4 Mission 9, Topic B, Lessons 5, 9
Statistics & Probability		
7.SP.1	Understand that statistics can be used to gain information about a population by examining a sample of the population.	Mission 8, Topic C, Lessons 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20
7.SP.2	Broaden statistical reasoning by using the GAISE model.	Mission 8, Topic C, Lessons 13, 14 Mission 8, Topic D, Lessons 15-17 Mission 8, Topic E, Lesson 20
7.SP.3	Describe and analyze distributions.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.SP.5	Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event; a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely; and a probability near 1 indicates a likely event.	Mission 8, Topic A, Lessons 2-6
7.SP.6	Approximate the probability of a chance event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability. For example, when rolling a number cube 600 times, predict that a 3 or 6 would be rolled roughly 200 times, but probably not exactly 200 times.	Mission 8, Topic A, Lessons 1, 3-6
7.SP.7	Develop a probability model and use it to find probabilities of events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy.	Mission 8, Topic A, Lessons 3-6 Mission 8, Topic C, Lesson 14 Mission 8, Topic E, Lesson 20
7.SP.8	Find probabilities of compound events using organized lists, tables, tree diagrams, and simulation.	Mission 8, Topic A, Lesson 6 Mission 8, Topic B, Lessons 7-10 Mission 8, Topic D, Lessons 15-19

8th Grade

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
8.NS.1	Know that real numbers are either rational or irrational. Understand informally that every number has a decimal expansion which is repeating, terminating, or is non-repeating and non-terminating.	Mission 8, Topic E, Lessons 14, 15
8.NS.2	Use rational approximations of irrational numbers to compare the size of irrational numbers, locate them approximately on a number line diagram, and estimate the value of expressions, e.g., π^2 . For example, by truncating the decimal expansion of $\sqrt{2}$, show that $\sqrt{2}$ is between 1 and 2, then between 1.4 and 1.5, and explain how to continue on to get better approximations.	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lessons 4, 5 Mission 8, Topic D, Lessons 12, 13
Expressions & Equations		
8.EE.1	Understand, explain, and apply the properties of integer exponents to generate equivalent numerical Expressions. For example, $32 \times 3^{-5} = 3^{-3} = 1/33 = 1/27$.	Mission 7, Topic B, Lessons 2-8 Mission 7, Topic C, Lessons 11, 14
8.EE.2	Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes. Know that $\sqrt{2}$ is irrational.	Mission 8, Topic B, Lessons 2-5 Mission 8, Topic C, Lesson 10 Mission 8, Topic D, Lessons 12, 13
8.EE.3	Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities and to express how many times as much one is than the other. For example, estimate the population of the United States as 3×10^8 ; and the population of the world as 7×10^9 ; and determine that the world population is more than 20 times larger.	Mission 7, Topic C, Lessons 9-12, 14 Mission 7, Topic D, Lesson 16

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.EE.4	Perform operations with numbers expressed in scientific notation, including problems where both decimal and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities, e.g., use millimeters per year for seafloor spreading. Interpret scientific notation that has been generated by technology.	Mission 7, Topic C, Lessons 10-15 Mission 7, Topic D, Lesson 16
8.EE.5	Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways. For example, compare a distance-time graph to a distance-time equation to determine which of two moving objects has greater speed.	Mission 3, Topic A, Lessons 2-4 Mission 3, Topic B, Lesson 6
8.EE.6	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b .	Mission 2, Topic C, Lessons 10-12 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lessons 10, 11 Mission C, Topic E, Lesson 14
8.EE.7	Solve linear equations in one variable.	Mission 4, Topic B, Lessons 3-9
8.EE.8	Analyze and solve pairs of simultaneous linear equations graphically.	Mission 3, Topic C, Lessons 13, 14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lessons 9-15 Mission 4, Topic D, Lesson 16
Functions		
8.F.1	Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. Function notation is not required in Grade 8.	Mission 5, Topic A, Lessons 1, 2 Mission 5, Topic B, Lessons 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
8.F.2	Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions). For example, given a linear function represented by a table of values and a linear function	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	represented by an algebraic expression, determine which function has the greater rate of change.	
8.F.3	Interpret the equation $y = mx + b$ as defining a linear function, whose graph is a straight line; give examples of functions that are not linear. For example, the function $A = s^2$ giving the area of a square as a function of its side length is not linear because its graph contains the points (1,1), (2,4) and (3,9), which are not on a straight line.	Mission 5, Topic B, Lessons 4, 7 Mission 5, Topic C, Lesson 8 Mission 5, Topic E, Lesson 18
8.F.4	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values.	Mission 5, Topic C, Lessons 8-10 Mission 5, Topic D, Lesson 11
8.F.5	Describe qualitatively the functional relationship between two quantities by analyzing a graph, e.g., where the function is increasing or decreasing, linear or nonlinear. Sketch a graph that exhibits the qualitative features of a function that has been described verbally.	Mission 5, Topic B, Lessons 5, 6 Mission 5, Topic C, Lesson 10
Geometry		
8.G.1	Verify experimentally the properties of rotations, reflections, and translations (include examples both with and without coordinates).	Mission 1, Topic A, Lessons 2-4, 6 Mission 1, Topic B, Lessons 7-10 Mission 1, Topic C, Lessons 11, 13, 14 Mission 1, Topic D, Lesson 14 Mission 3, Topic B, Lesson 8
8.G.2	Understand that a two-dimensional figure is congruent to another if the second can be obtained from the first by a sequence of rotations, reflections, and translations; given two congruent figures, describe a sequence that exhibits the congruence between them. (Include examples both with and without coordinates.)	Mission 1, Topic C, Lessons 11-13 Mission 1, Topic D, Lesson 15 Mission 2, Topic B, Lessons 6, 7

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.G.3	Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates.	Mission 2, Topic B, Lesson 7 Mission 1, Topic A, Lessons 5, 6 Mission 2, Topic A, Lessons 4, 5 Mission 2, Topic C, Lesson 12
8.G.4	Understand that a two-dimensional figure is similar to another if the second can be obtained from the first by a sequence of rotations, reflections, translations, and dilations; given two similar two-dimensional figures, describe a sequence that exhibits the similarity between them. (Include examples both with and without coordinates.)	Mission 2, Topic B, Lessons 6, 7, 9
8.G.5	Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles. For example, arrange three copies of the same triangle so that the sum of the three angles appears to form a line, and give an argument in terms of transversals why this is so.	Mission 1, Topic D, Lessons 14-16 Mission 2, Topic B, Lesson 8 Mission 2, Topic D, Lesson 13 Mission 9, Topic A, Lesson 2
8.G.6	Analyze and justify an informal proof of the Pythagorean Theorem and its converse.	Mission 8, Topic C, Lessons 7, 9
8.G.7	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions.	Mission 8, Topic C, Lessons 6-8, 10
8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C, Lesson 11
8.G.9	Solve real-world and mathematical problems involving volumes of cones, cylinders, and spheres.	Mission 5, Topic D, Lessons 13-16 Mission 5, Topic E, Lessons 17-21 Mission 5, Topic F, Lesson 22
Statistics & Probability		
8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering; outliers; positive, negative, or no	Mission 6, Topic A, Lessons 1, 2 Mission 6, Topic B, Lessons 3-8

OHIO'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
	association; and linear association and nonlinear association.	
8.SP.2	Understand that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line.	Mission 6, Topic B, Lessons 4-6, 8
8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercept. For example, in a linear model for a biology experiment, interpret a slope of 1.5 cm/hr as meaning that an additional hour of sunlight each day is associated with an additional 1.5 cm in mature plant height.	Mission 6, Topic B, Lessons 6, 8
8.SP.4	Understand that patterns of association can also be seen in bivariate categorical data by displaying frequencies and relative frequencies in a two-way table. Construct and interpret a two-way table summarizing data on two categorical variables collected from the same subjects. Use relative frequencies calculated for rows or columns to describe possible association between the two variables. For example, collect data from students in your class on whether or not they have a curfew on school nights and whether or not they have assigned chores at home. Is there evidence that those who have a curfew also tend to have chores?	Mission 6, Topic B, Lessons 9, 10